

Yixuan Wang

Ph.D. Student, University of Florida · Expected May 2027 · Gainesville, FL

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RESEARCH PROFILE

Machine learning researcher using nonlinear control, dynamical systems, and geometry to understand generative models and LLMs, improve model structure, and develop new algorithms. First-author papers at ICML (Spotlight), NeurIPS, IEEE CDC, and IFAC World Congress. Seeking research scientist roles starting summer 2027.

SELECTED RESEARCH CONTRIBUTIONS

- **Effective Model Pruning (ICML 2026 Spotlight).** Derived adaptive sparsity levels from component importance scores via effective sample size, with a provable loss-change bound; implemented and validated across MLPs, CNNs, Transformers/LLMs, and KANs.
- **Riemannian Lyapunov framework (NeurIPS 2026).** Formulated optimization as closed-loop control on Riemannian manifolds, recovering classical optimizers and deriving new update laws; implemented and benchmarked on ImageNet-scale ResNet and ViT classification.
- **Generative modeling as distributional control.** Developed a Lyapunov perspective on energy-based generative models in which sampling is a control problem, yielding a principled finite-step stopping rule for Langevin sampling.
- **Bayesian intent inference and privacy.** Built a Rao-Blackwellized particle filter for online goal inference from trajectories, and control policies that jointly regulate physical motion and an observer's belief, with formal bounds on goal information disclosure (IFAC 2026, IEEE CDC 2026).
- **Research proposal development.** Led the development and writing of 1-2 research aims per proposal, including technical approaches, preliminary experiments, and research plans, for advisor-led proposals funded by NSF, AFOSR, and AFRL.

TECHNICAL SKILLS

Machine Learning	PyTorch; generative models (energy-based, diffusion, flow matching); reinforcement learning; model compression
ML Systems	Multi-GPU distributed training (DDP); Slurm-based HPC workflows (UF HiPerGator)
Programming	Python, C/C++, MATLAB / Simulink, Git, LaTeX

SELECTED PUBLICATIONS

- Effective Model Pruning: Measuring the Redundancy of Model Components — *ICML 2026* Spotlight
- Riemannian Lyapunov Framework: Optimization as Closed-Loop Control on Riemannian Manifolds — *NeurIPS 2026*
- Off-Dynamics Reinforcement Learning via Domain Adaptation and Reward Augmented Imitation — *NeurIPS 2024*
- Quantifying Trade-Offs Between Stability and Goal-Obfuscation — *IEEE CDC 2026*
- Goal Inference with Rao-Blackwellized Particle Filters — *IFAC World Congress 2026*

PROFESSIONAL SERVICE

Conference Reviewer: ICLR, ICML, NeurIPS (since 2025).

EDUCATION

University of Florida	— Ph.D., Mechanical & Aerospace Engineering	2024 – 2027 (exp.)
Johns Hopkins University	— M.S., Computer Science	2023 – 2024
University of Florida	— M.S., Mechanical Engineering	2020 – 2022
Tongji University	— B.S., Mechanical Design, Manufacture and Automation	2015 – 2020